SHENZHEN UNIVERSITY

COURSE

Data Analytics in Marketing
Shenzhen University

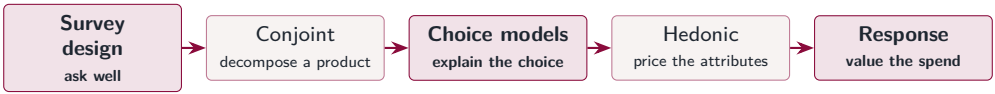
Chapter 15 — Big Data in Marketing
Survey design, conjoint, choice models, hedonic regression and response models

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Five Ways Marketing Data Becomes a Number



The five are a sequence, not a menu: survey design decides what you can ask, conjoint turns product attributes into part-worths, choice models turn part-worths into shares, hedonic regression prices attributes that already exist in the market, and response models connect marketing spend to outcomes.

Source: The guideline of the module reading: survey design, conjoint analysis, choice modelling, hedonic regression, response models.

Learning Objectives

After this lecture you should be able to:

1. design a survey that supports inference — sample, instrument, mode — and explain the main sources of survey error;
2. run a conjoint study end to end: study type, attributes, levels, fractional-factorial design, and interpretation of part-worths;
3. explain random utility, write the logit choice probability, and say what the IIA assumption rules out;
4. estimate a hedonic price equation and use it to build a quality-adjusted price index;
5. read a response model: concavity, elasticity, and how to allocate a budget across channels;
6. judge when each method is the right tool — and when the data cannot support it.

Source: All five methods are used in the course project; the rubric rewards the justification, not the software.

Where These Methods Sit



The gap L5 leaves

a predictive model tells you what will happen, not why — and not what happens if you change the price, the pack or the message.

What L6 adds

structural quantities: part-worths, elasticities, willingness to pay — numbers a manager can act on and a finance team can challenge.

Source: The distinction matters: a churn score ranks customers, a choice model tells you which lever moves them.

What Survey Methodology Is

Survey methodology is the study of how to ask: sampling units from a population, collecting data, constructing questionnaires, and improving the number and accuracy of responses.

It is a method, not a form

A survey is an instrument for inference about a population. The questionnaire is only its visible part; the sample design and the fieldwork decide whether the numbers mean anything.

Where it is used

public opinion, public health, market research, government statistics and censuses — any quantitative research that needs to describe a population from a subset of it.

Note that a census has no sampling error but still uses questionnaires, interviewers and non-response follow-up: it is survey methodology without the sample.

Inference and Representativeness

Every survey-based number is an estimate, and its credibility rests on one question: is the sample representative of the population you claim to describe?

The population must be named

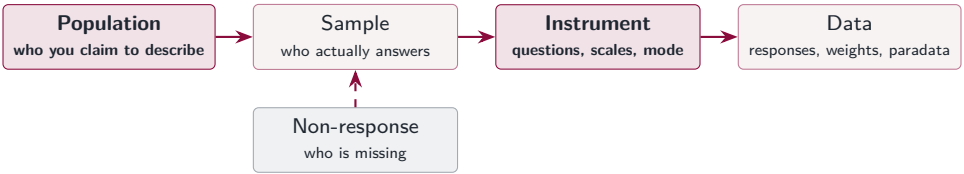
"Chinese consumers" is not a population. "Adults 25–45 in tier-1 cities who bought skincare online in the last 90 days" is. The narrower the definition, the more defensible the inference — and the smaller the market you can speak about.

Respondents are not the population

People answer as individuals, as households, as employees, or on behalf of an organisation. Decide which unit your question is about before you weight the answers.

Practical consequence: an online panel of 2,000 respondents can be less informative than 200 carefully sampled ones, because who answers dominates how many answer.

Anatomy of a Survey



Three components are present in every survey: a sample (or a census), a method of collection, and individual items that become analysable data. Two of the three are invisible in the dataset you finally receive — which is why the documentation matters.

Source: A single survey may also be only one wave of a panel; then the wave design is a fourth component.

Four Kinds of Survey Question

What people say they prefer

Preferences — which candidate, which pack, which feature. Easy to ask, weakly linked to behaviour, and sensitive to how the options are framed.

What people did

Behaviour — purchases, usage, channel. Recall error is large and systematically biased towards recent, salient and socially desirable events.

What people believe

Opinions — agree/disagree, importance ratings, satisfaction. Useful for diagnosis, poor for forecasting.

What is true about them

Facts — income, age, household size. The easiest to ask and the most often refused or misreported.

Design rule: ask behaviour where you can observe it, and treat stated preference as an input to a model (conjoint, choice experiment) rather than as a forecast.

Survey Error and Cost

Survey methodology studies the errors that arise from sample design, collection instruments, statistical adjustment, data processing and final analysis.

The four error families

- **coverage** — the frame misses part of the population;
- **sampling** — the luck of the draw;
- **non-response** — who declines, and why;
- **measurement** — question wording, mode, interviewer.

The budget trade-off

Quality is bought: longer fieldwork, incentives, and interviewer training reduce non-response error but cost money. The design question is always which error to buy down at a fixed budget.

Sampling error is the only one you can compute from the data you have — which is why it gets reported and the other three get ignored.

Seven Methodological Decisions

1. **Identify and select** potential sample members — and write down the frame you drew them from.
2. **Contact** sampled individuals, and collect data from those who are hard to reach or reluctant; budget for repeated attempts.
3. **Evaluate and test** the questions — cognitive interviews, pilots, wording experiments.
4. **Select the mode** for posing questions and collecting responses (next slide).
5. **Train and supervise** interviewers, where interviewers are involved; their behaviour is a source of variance.
6. **Check the data files** for accuracy and internal consistency before analysis.
7. **Adjust the estimates** to correct for identified errors — weights, calibration, imputation.

Source: These seven decisions are where survey quality is won or lost; software cannot rescue a bad frame.

Modes of Administration

Mode	Coverage	Cost per response	Main bias
Telephone	high, but falling	low–medium	excludes the young; short attention
Mail (post)	good for older, rural	low	slow; literacy and language
Online panel	skewed to connected	very low	professional respondents, speeders
In-home	excellent	very high	interviewer effects, social desirability
Mall / street intercept	urban, mobile	medium	convenience sample, short interviews
Hybrid	design-dependent	medium	mixing modes creates mode effects

The choice is driven by five factors: cost, coverage of the target population, flexibility of questioning, willingness to participate, and accuracy of the response.

Mode Effects

Different modes are not neutral pipes: they change how people answer, not only who answers.

Typical effects

- telephone: shorter, more positive, more "don't know";
- self-administered: more honest on sensitive items;
- interviewer-present: more socially desirable answers;
- mobile: higher break-off, shorter open text.

What to do about it

- keep the mode constant across the comparison you care about;
- if you must mix, test for mode effects explicitly;
- prefer self-administered for sensitive behaviour;
- keep the instrument short on mobile.

Source: If a survey changes mode mid-fieldwork, treat the change as a break in the series, not a detail.

A Survey Design Checklist

1. **Population and frame.** Who exactly, drawn from what list, with what known gaps?
2. **Unit of analysis.** Individual, household, firm, listing?
3. **Sample size and power.** What difference must the survey be able to detect?
4. **Instrument.** Are the questions tested, unambiguous, and free of double-barrelled phrasing?
5. **Mode.** Does it reach this population, and does it bias the sensitive items?
6. **Fieldwork.** Attempts, incentives, quotas, interviewer training, quality checks.
7. **Weighting.** Which known population totals will you calibrate to?
8. **Documentation.** Questionnaire, codebook, fieldwork dates, response rate, weighting scheme.

Source: This list is also the minimum documentation for a survey-based project submission.

What Conjoint Analysis Does

Conjoint analysis is a survey-based decomposition: respondents evaluate complete product profiles, and the analyst works backwards to the value of each attribute.

The logic

A product is a bundle. Ask people to choose between bundles, not to rate attributes in isolation — because in isolation everything is important.

What comes out

part-worth utilities for each level, and from them the implicit price of a feature, the trade-off between attributes, and simulated market shares.

Developed by Paul Green at Wharton from mathematical psychology; now standard in product design, pricing, advertising appeal and service configuration.

Source: The method is also called multiattribute compositional modelling, stated-preference research or trade-off analysis.

Part-Worths and Market Models

Each profile j has attributes x_{jk}; utility is additive in part-worths:

$$U_j = \sum_k \sum_l \beta_{kl} \mathbf{1}\{x_{jk} = l\} + \varepsilon_j$$

Choice is probabilistic, so shares follow a logit form:

$$s_j = \frac{\exp(U_j)}{\sum_{i \in J} \exp(U_i)}$$

What you can then ask

- what share does a new configuration win?
- what price premium does feature X support?
- which competitor loses most when we launch?

The share formula is the bridge to the next section: conjoint gives you U_j, and the choice model turns utilities into shares. Both rely on the same random-utility foundation.

Source: Adding an outside option ("none of these") is essential if you want realistic shares; without it the model must allocate 100% across the profiles shown.

Step 1: Choose the Study Type

Ranking-based	Rating-based	Choice-based
Respondents rank a set of profiles. Rich information per respondent, but tiring beyond $\sim$15 profiles.	Rate each profile on a scale. Easy to administer, but scale use varies across respondents and cultures.	Choose one profile from each set (CBC). Closest to a real purchase, handles "none", needs more respondents.

For an international study, the third column has a practical advantage: a choice task needs no translation of a rating scale, which removes a well-known source of cross-country incomparability.

Source: Scale-use bias is one reason rating scales travel badly across markets; choice tasks are more portable.

Step 2: Identify Attributes

Attributes must satisfy six conditions:

- **relevant** to a managerial decision — otherwise the study answers nothing;
- **varying** in real life — a constant attribute cannot be traded off;
- **influential** on preference — test with a pilot, do not assume;
- **clearly defined** and communicable in a few words;
- **not strongly correlated** with each other (price and brand are the accepted exception);
- **at least two levels** each.

The usual failure is the third condition: studies that include attributes nobody cares about, which dilute the design and lengthen the interview without adding information.

Source: Six to eight attributes is a practical ceiling for a choice-based study; beyond that, respondents simplify.

Step 3: Specify Levels

Levels are the values each attribute takes, and they carry the entire informational content of the design. They must be:

- **unambiguous** — "medium" is not a level; "4.5–5.0 inch" is;
- **mutually exclusive** — no overlap between adjacent levels;
- **realistic** — inside the range the market actually offers;
- **balanced** — no level so extreme that it dominates every choice.

Range matters

Part-worths are estimated relative to the levels you showed. If your price range is ¥2,000–¥3,500, the model cannot tell you anything about a ¥1,500 product — and the estimated price sensitivity will be too flat.

Design the range to bracket the decision you are actually considering, including the option you expect to lose.

Step 4: Design the Questionnaire

Combinations explode: with 5 attributes at 3 levels each, the full factorial is

$$3^5 = 243 \text{ profiles},$$

far more than any respondent will evaluate. So we use a fractional factorial design — a carefully chosen subset that keeps the main effects estimable.

What the design buys

an orthogonal subset (commonly 16–27 profiles), blocked into 8–12 choice tasks per respondent, with enough overlap to estimate the model.

What it costs

some interactions are confounded with main effects and cannot be recovered. Decide in advance which interactions you are willing to give up.

Source: Software: `AlgDesign`/`conjoint` in R, Sawtooth, or a hand-built orthogonal array; always report the design used.

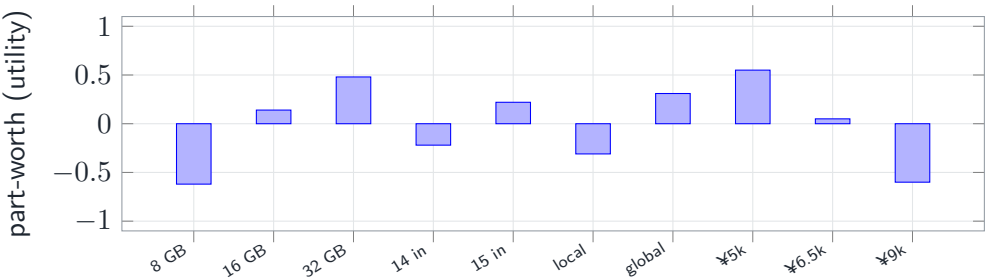
Worked Example: Laptop Profiles

Profile	Price	RAM	Screen	Brand	Choice share
1	¥4,999	8 GB	14″	local	18%
2	¥4,999	16 GB	14″	global	34%
3	¥6,499	16 GB	15″	global	21%
4	¥6,499	8 GB	15″	local	9%
5	¥8,999	16 GB	15″	global	12%
6	¥8,999	32 GB	15″	global	6%

From six profiles you can already read the trade-off: doubling RAM at a constant price adds more share than a one-inch larger screen, and the brand effect is worth roughly one price step. A real study uses 16–27 profiles and estimates these effects jointly.

Source: Illustrative numbers, consistent with a linear price effect and additive part-worths.

Reading Part-Worths



Part-worths are on an arbitrary scale: only differences within an attribute are meaningful. The range of part-worths within an attribute measures how much that attribute matters; the ratio between a feature's part-worth and price's part-worth is the implicit price of the feature.

Advantages

- estimates the psychological trade-offs consumers make when judging attributes together;
- measures preferences at the individual level, enabling segment-level analysis;
- uncovers drivers respondents cannot articulate — including ones they are unaware of;
- mimics a realistic choice or shopping task, especially in choice-based form;
- can use physical objects, packaging or screenshots;
- models interactions if the design allows;
- supports needs-based segmentation when heterogeneity is modelled.

The strongest argument for conjoint over direct questioning: people are poor at stating the weight they put on an attribute, but reasonably good at choosing between bundles.

Disadvantages

- design is genuinely complex — a bad design cannot be rescued in analysis;
- with too many features, respondents adopt simplification strategies (ignore an attribute, always pick the cheapest);
- poor for positioning research: perceptions of actual features are not the same as perceptions of underlying dimensions;
- emotional features can be over-valued and concrete ones under-valued in weak designs;
- ignores purchase quantity unless volumetric extensions are used.

Practical mitigation: keep the task short, randomise order, include an outside option, screen out speeders, and report the design and the screening rules alongside the estimates.

What Choice Modelling Is

Choice modelling estimates the decision process of an individual or segment, using revealed preferences (what people actually bought) or stated preferences (what they chose in an experiment).

The core move

Observe discrete choices — A over B, B over A, B and C over A — and infer where the alternatives sit on a latent scale, usually utility.

What it is good for

willingness to pay for quality improvements in multiple dimensions at once — the question a simple price elasticity cannot answer.

The models come from econometrics, marketing and sociometrics: utility maximisation, consumer theory, and a family of identification strategies that differ in what they assume about substitution.

Discrete versus Continuous Choice

Continuous: "how much?"

Standard consumption models treat quantity as continuous, so calculus applies: first-order conditions give the optimum, and demand is estimated by regression.

Discrete: "which one?"

The outcome is a choice among a finite set, so the optimum is not characterised by standard first-order conditions. The object of interest is a probability, not a quantity.

Discrete analysis is the right frame whenever the decision is which — which car, which university, which transport mode, which pack size — rather than how many units. `logit` and `probit` are the empirical workhorses; `logistic` regression is the binary special case.

Random Utility and McFadden

The decision maker i facing alternatives j obtains utility

$$U_{ij} = V_{ij} + \varepsilon_{ij}, \qquad V_{ij} = \beta' x_{ij},$$

where V is the observed part and ε the analyst's ignorance. Choosing the best option gives a probability, because ε is unobserved.

The Nobel connection

Daniel McFadden developed the theoretical basis of discrete choice and shared the 2000 Nobel Memorial Prize. His contribution was to make the probabilistic choice model estimable and interpretable — including the conditional logit.

Everything that follows is a statement about the distribution of ε. Assume a Gumbel distribution and you get logit; assume normal and you get probit; allow correlation and you get the nested, GEV and mixed families.

Two Kinds of Data

Consumer characteristics

ID	Job	Age	Income
1	Student	20	0
2	Manager	45	80,000
3	Doctor	35	120,000
4	Driver	42	60,000

Product characteristics

ID	Price	GHz	MP	g
1	1,900	2.5	12	250
2	2,800	3.0	20	200
3	3,500	4.0	40	280
4	2,400	3.0	20	230

Choice models need both tables, joined by the choice record. The consumer table supplies taste heterogeneity; the product table supplies the attributes whose coefficients are the object of interest.

Individual versus Aggregate Demand Data

Individual demand

Consumer	Product	Job	Age	Price
1	3	Student	20	\$90
2	4	Worker	45	\$80
3	1	—	—	—

each row = one purchase by one person

Aggregate demand

Share	ID	Colour	Type	Price
25%	1	2	1	\$90
10%	2	3	1	\$80
20%	3	3	—	—

each row = a market share for a product

The distinction determines the estimator: individual data allow a choice model with taste heterogeneity; aggregate data give a market-share model and identify less about individuals.

An Applied Example: NEV Policy and Conventional Cars

Variable	N	Mean	S.D.	Min / Max
Engine displacement (L)	383	1.59	0.91	0 / 6
Retail volume (units)	383	73.6	171.6	10 / 2,050
MSRP (¥10k)	383	31.1	33.4	3.3 / 328.5
Fuel consumption (L/100km)	383	5.25	3.41	0 / 23.1
Motor power (kW)	383	45.7	74.9	0 / 510
National subsidy (¥10k)	383	0.26	0.44	0 / 1.39
Shenzhen subsidy (¥10k)	383	0.42	0.78	0 / 2.0
SUV (dummy)	387	0.51	0.50	0 / 1
Joint venture (dummy)	387	0.48	0.50	0 / 1
Domestic brand (dummy)	387	0.33	0.47	0 / 1

This is the descriptive table behind a study of how new-energy-vehicle subsidies shifted demand for conventional cars: the outcome is a discrete choice among models, and the subsidies enter as attributes that differ by city and by model.

Source: Illustrative summary statistics reconstructed for teaching; the identification argument is what matters here.

Binomial and Multinomial

Binomial (dichotomous)

Two alternatives: buy or not, enter or not, churn or stay.

$$\Pr(y_i = 1) = \Lambda(x_i'\beta)$$

Estimated by logit or probit; this is ordinary binary-choice regression.

Multinomial (polytomous)

Three or more alternatives, further split by whether unobserved factors are allowed to correlate across alternatives:

- independent (standard logit);
- correlated (nested logit, GEV, probit, mixed logit).

The practical difference is substitution: the multinomial model must say what a customer buys *instead* when your product disappears — and that is where the assumption bites.

The Choice Set

The choice set is the set of alternatives actually available to the person. Three requirements:

- **Collectively exhaustive** — every possible alternative is in the set (including "none" or "other"), so the person necessarily chooses one;
- **Mutually exclusive** — choosing one excludes the others, so exactly one is chosen;
- **Finite** — a countable number of alternatives. This is what distinguishes discrete choice from regression, where y can take infinitely many values.

Choice sets differ across people: the Scion was not sold in Canada in 2009, so Canadian buyers faced a different set from American buyers. Ignoring that difference biases the estimates — it is a coverage problem in the sense of the survey section.

Choice Probabilities: Logit

With Gumbel-distributed errors, the probability that i chooses j has the closed form

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k \in J_i} \exp(V_{ik})}, \qquad V_{ij} = \beta' x_{ij}.$$

Only differences in V matter, so the scale of utility is unidentified and one alternative must be normalised.

Why the closed form matters

It makes estimation fast, shares computable by hand, and elasticities available analytically:

$$\frac{\partial \ln P_{ij}}{\partial x_{ijk}} = \beta_k \big(1 - P_{ij}\big).$$

The same formula is what turns conjoint part-worths into market shares — the two sections share one engine.

Consumer Utility

Utility has an observed part and an unobserved part, and the decomposition is not cosmetic — it determines which models are estimable:

$$U_{ij} = \underbrace{\sum_k \beta_k x_{ijk}}_{\text{observed } V_{ij}} + \underbrace{\varepsilon_{ij}}_{\text{unobserved}} .$$

What ε absorbs

taste variation the analyst cannot see, measurement error, and attributes the researcher did not include. Its assumed distribution is the model's main assumption.

What the coefficients mean

β_k is the marginal utility of attribute k. Ratios $-\beta_k/\beta_{\text{price}}$ give willingness to pay, which is usually the number the business wants.

Binary Choice: A Worked Example

A retailer wants to know whether a free-returns badge raises conversion. Each visitor chooses between two otherwise identical offers, one with the badge:

$$V_{\text{badge}} = \beta_0 + \beta_1 \cdot 1, \qquad V_{\text{no badge}} = \beta_0, \qquad \Pr(\text{badge}) = \frac{e^{\beta_1}}{1 + e^{\beta_1}}.$$

Reading the estimate

If $\hat{\beta}_1 = 0.35$, the odds of choosing the badged offer rise by $e^{0.35} \approx 1.42$, i.e. about 42%. The probability effect is smaller and depends on the baseline.

The trap

A single binary choice identifies one parameter. It cannot tell you the value of the badge separately from everything else that differed between the two offers.

Multinomial Logit

With more than two alternatives, the same formula applies to the whole choice set, and the model delivers a full matrix of cross-elasticities:

$$P_{ij} = \frac{e^{V_{ij}}}{\sum_{k=1}^{J} e^{V_{ik}}}, \qquad \eta_{j \to k} = \frac{\partial \ln P_{ij}}{\partial \ln p_k} = -p_k \, \beta_{\text{price}} \quad (k \neq j).$$

What is attractive

closed-form shares, easy simulation of new products, and elasticities that can be reported per competitor.

What is uncomfortable

the cross-elasticity above is the same for every competitor: a price cut by one brand draws proportionally from all others.

That property is not a modelling convenience — it is an assumption about consumer behaviour, and it has a name.

The IIA Problem

Independence of Irrelevant Alternatives: the ratio of probabilities of two alternatives does not depend on what else is in the choice set.

$$\frac{P_{ij}}{P_{ik}} = \frac{e^{V_{ij}}}{e^{V_{ik}}} \quad \text{— independent of } J_i.$$

The red bus / blue bus example

With a car and a red bus, suppose shares are 50/50. Add a blue bus that is identical to the red one except in colour. Logit predicts the car's share falls to one third — as if a new bus were as different from the red bus as the car is.

Why it matters

Real substitution is local: the blue bus steals mostly from the red bus. Logit overpredicts the loss of distant alternatives — and therefore mis-states the effect of a new entrant.

Models That Relax IIA

Model	What it allows
Nested logit	partitions the choice set into "nests"; correlation within a nest, independence across
Cross-nested logit (CNL)	an alternative may belong to more than one nest
C-logit	correlation captured through a commonality factor between alternatives
Paired combinatorial logit	pairwise correlation; suited to route and mode choice
Generalised extreme value (GEV)	the general family containing logit and nested logit
Conditional probit	full covariance across alternatives via a joint normal distribution
Mixed logit	any correlation and substitution pattern; random coefficients, panel structure

Practical guidance: start with logit to see whether IIA is plausible for your category; if a new entrant's share is being over-predicted, move to nested or mixed logit and report both.

Source: The nested-logit structure is the standard fix in marketing; mixed logit is the modern default when individual data are available.

What Hedonic Regression Is

Hedonic regression is a revealed-preference method: it decomposes a good into its characteristics and estimates the contributory value of each one from market prices.

The two requirements

- the composite good can be reduced to constituent parts;
- those parts are valued by the market — there is a price for them.

How it is estimated

usually by regression on an attribute vector (dummy or continuous per characteristic), which accommodates non-linearity and interaction.

It is the market's implicit conjoint: instead of asking people to trade off attributes, you read the trade-offs off the prices they actually paid.

Real-Estate Valuation

Housing is the canonical case: every unit is different, so "the price of housing" is meaningless without a quality adjustment.

The decomposition

A house is treated as a bundle: bedrooms, plot size, floor, age, distance to the city centre, school district, transport. The regression prices each attribute separately.

What the estimates buy

a quality-adjusted price index — comparable across cities and over time — and a valuation for a property with no transaction of its own.

In the United States, mass appraisal with hedonic models is governed by the USPAP standards; the model is used for tax assessment, not only for research.

The Hedonic Equation

The workhorse specification is log-linear, so coefficients read as percentage effects:

$$\ln P_i = \alpha + \sum_{k=1}^{K} \beta_k X_{ik} + \varepsilon_i, \qquad \hat{\beta}_k \approx \frac{\Delta P}{P} \text{ per unit of } X_k.$$

A semi-log form allows diminishing returns; a fully log form gives elasticities.

Building the index

1. estimate $\hat{\beta}$ on a base period;
2. hold the attribute vector fixed;
3. let only prices move — the residual change is the pure price change.

This is exactly how statistical agencies handle quality change in the CPI: without it, a price rise caused by a better product would be recorded as inflation.

Source: The same logic underpins quality-adjusted price indices for electronics, cars and services.

Where the Data Come From

Listing platforms

Lianjia / Beike transaction and listing records: address, area, floor, age, layout, price, date.

Short-let platforms

Airbnb: nightly price, capacity, amenities, ratings, distance to centre, seasonality.

Travel platforms

Booking / Ctrip: hotel attributes, star rating, review scores, occupancy proxies, cancellation policy.

Two data-quality warnings that decide the study. First, listed price is not transaction price — negotiate, and the gap is itself informative. Second, platform data are a selected sample: properties not listed are missing, and that selection is correlated with quality.

Source: Scraped platform data must be documented as to date, coverage and terms of use.

Beyond Real Estate

Price statistics

consumer price indices with quality adjustment; hedonic deflators for ICT goods, where quality moves faster than price.

Assessment and litigation

mass appraisal for property tax; damages estimation; transfer-pricing comparables; sector-wide valuation where transactions are sparse.

A marketing application

the same equation prices product attributes in any category with enough variation — a cuisine set whose pans differ in material and coating, a laptop differing in RAM and screen, a hotel differing in location and breakfast. The coefficient on an attribute is what the market pays for it.

Source: Appraisal practice treats hedonic regression as a statistically robust form of the sales-comparison approach.

What a Response Model Is

A response model is a mathematical model of the relationship between marketing investment and outcome, built to allocate resources rather than to explain the world.

The assumption

past outcomes relate to future outcomes — reasonable most of the time, and wrong in structural breaks (a recession, a platform rule change, a competitor exit).

The output

a curve, not a point: how much outcome you get for the next yuan, and where that return turns flat.

This is the marketing-mix modelling tradition: aggregate time-series or panel data, advertising and promotion as inputs, sales or share as output.

Three Questions a Response Model Answers

Shape	Comparison	Competition
Is the relationship linear, concave, or S-shaped? Where does saturation begin?	Which marketing-mix instrument produces more outcome per unit of spend?	How much of our outcome is explained by our effort, and how much by competitors' effort?

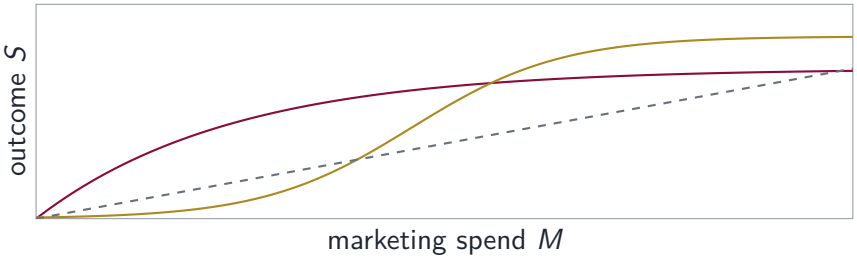
The competitive question is the one most often skipped — and the one that most often explains a disappointing result: a campaign that worked, in a category where everyone spent at once, can show no measurable effect at the brand level.

Four Insights

1. **The shape is usually concave.** Outcomes increase with marketing resources, but at a diminishing rate — which is why the marginal, not the average, return governs the decision.
2. **Elasticity is directly readable.** The model tells you the percentage change in outcome for a 1% increase in effort — the number a planning meeting actually argues about.
3. **Relative impact supports allocation.** Comparing coefficients across instruments allows a budget to be split in proportion to effectiveness, not to history.
4. **Competitive effects can be separated.** Controlling for rivals' spending avoids attributing their effort to your own, which is the classic omitted-variable error in marketing-mix models.

Source: The four insights follow the marketing-response literature (Hanssens, Parsons & Schultz); the caveat about breaks is ours.

The Concave Response Curve



The dashed line is the assumption most budget spreadsheets make. The gap between it and the red curve at high spend is the money that a response model saves — and at low spend, it is the money that a linear model wastes.

Budget Allocation from Response Models

With concave responses $S_i = a_i M_i^{b_i}$, the optimal split equalises the marginal return per yuan across instruments:

$$\frac{\partial S_1}{\partial M_1} = \frac{\partial S_2}{\partial M_2} = \cdots = \lambda, \qquad M_i^{*} \propto \left(\frac{a_i b_i}{\lambda}\right)^{\frac{1}{1-b_i}}.$$

What this implies

channels with a high b (slowly saturating) deserve more budget; channels already saturated get less, even if their average return looks good.

What it requires

credible estimates of a_i and b_i per channel — which usually means either an experiment or a long enough history with variation in spend.

Without variation in spend, the curve is not identified: a channel that has always received the same budget gives you one point, and one point is not a curve.

Reading These Models Critically

1. **What is the identifying variation?** For conjoint: the design. For choice models: the variation in attributes. For hedonic: the variation in characteristics at a point in time. For response: the variation in spend.
2. **What is assumed about substitution?** IIA, nested structure, or full covariance — say which, and test the implication you care about.
3. **Is the counterfactual inside the data range?** A model estimated on ¥2,000–3,500 products cannot price a ¥1,500 one.
4. **Are the units and the scale meaningful?** Part-worths are relative; elasticities are not.
5. **What would falsify it?** A validation sample, an experiment, or a hold-out period.
6. **What decision does it change?** If the answer does not move a price, a product or a budget, it is description, not analysis.

Source: This is the checklist the project rubric applies to the modelling section.

Summary

1. **Survey design** is inference infrastructure: population, frame, instrument, mode and weighting decide whether the numbers mean anything.
2. **Conjoint** decomposes a product into part-worths by making respondents choose between bundles; it gives the implicit price of features and the trade-offs between attributes.
3. **Choice models** turn utilities into shares through the logit form; the IIA assumption is the fault line, and nested or mixed logit is the usual repair.
4. **Hedonic regression** reads attribute prices off market data and makes quality-adjusted price indexation possible.
5. **Response models** connect spend to outcome, are usually concave, and exist to allocate budgets — which requires variation in spend to be identified at all.
6. Across all five: the design determines what can be learned, and no estimation method can recover what the design did not vary.

Source: Next: the project uses at least one of these five methods, with the design and the limitation stated explicitly.

Thank You

Data, code and the project brief are on the course site.